

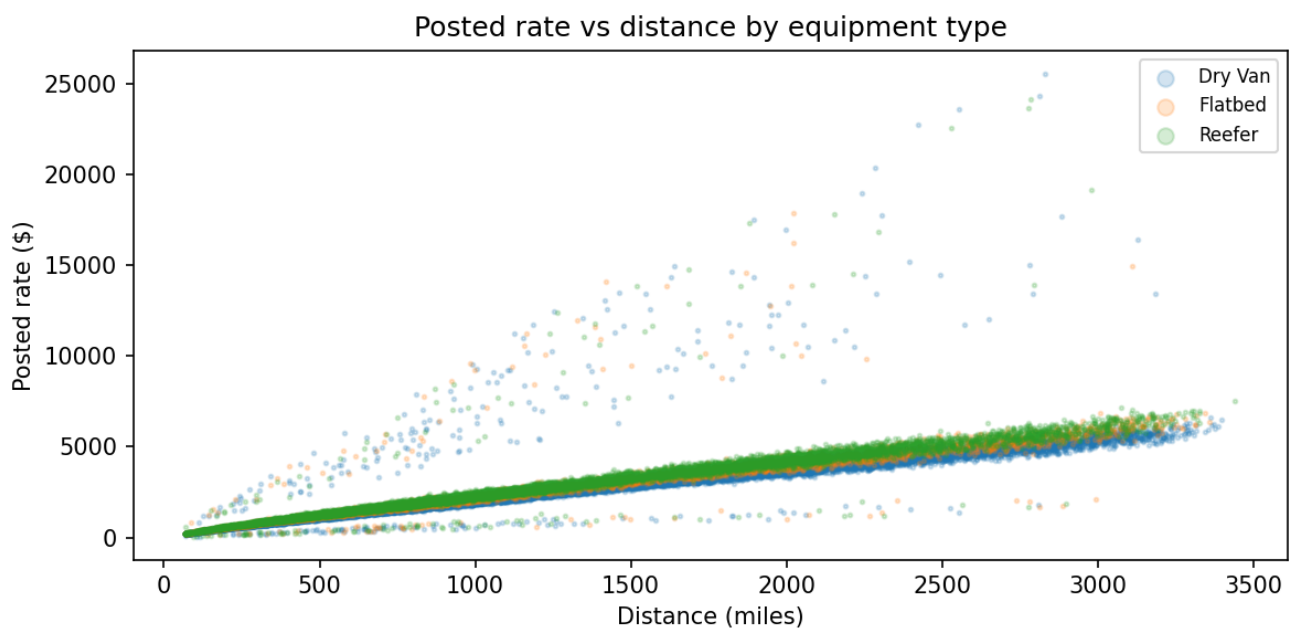
Freight Rate Prediction

Goal: predict the posted rate for 12,000 truckload freight loads dated November-December 2025 (data/validation.csv), given 48,000 labeled loads from January-October 2025 (data/train_test.csv), plus a fixed daily December 2025 rate chart for a reference lane (Lexington to Fort Wayne, 360 mi, Dry Van, 32,000 lb).

Solution: a LightGBM gradient-boosted tree ensemble trained on $\log(\text{posted_rate})$ with an L1 objective, validated with a strict time-based holdout (train Jan-Sep, validate Oct), then refit on all ten months across 5 seeds whose predictions are averaged.

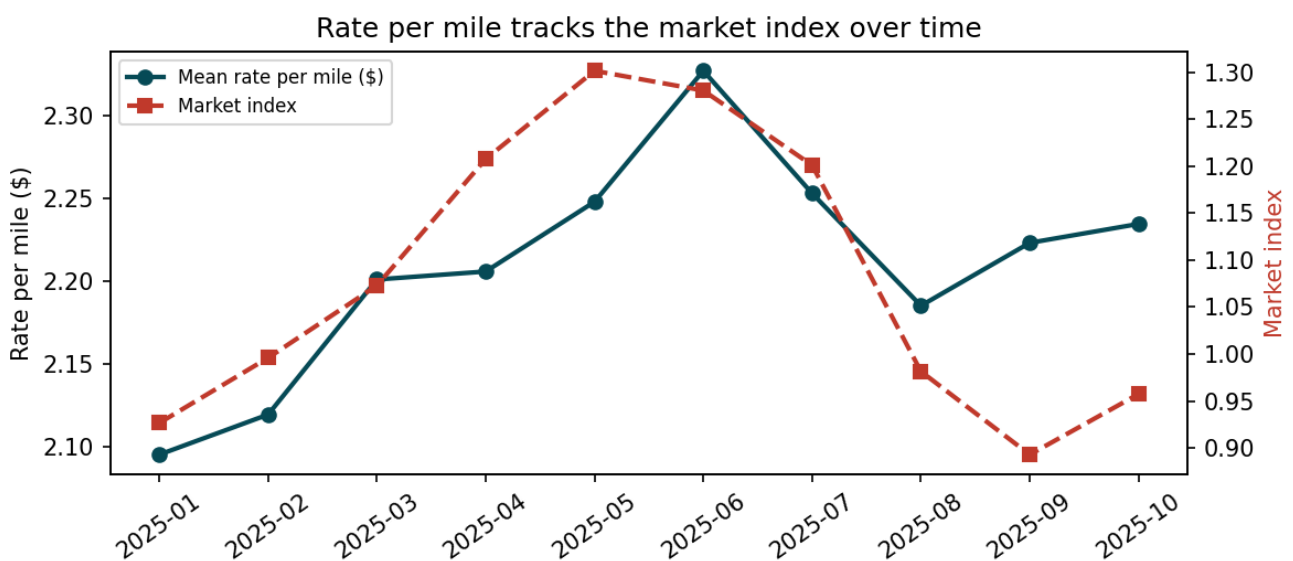
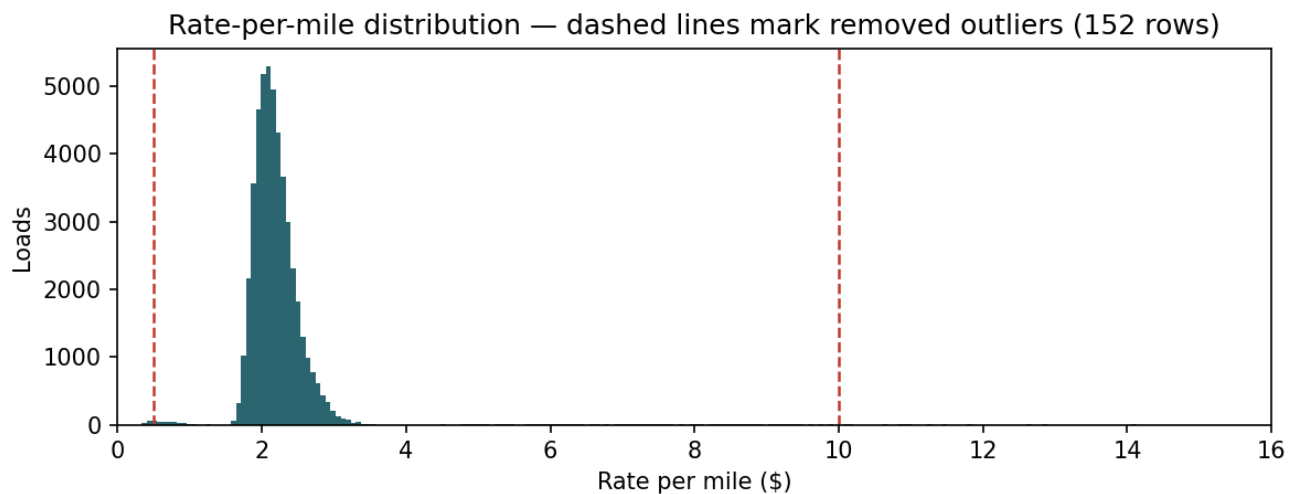
Key findings from data exploration

- Distance dominates: correlation with `posted_rate` is 0.91, and rate per mile falls as distance grows (long hauls price lower per mile).
- Rates are multiplicative - rate per mile is roughly log-normal with a median of \$2.15/mi - so the model predicts $\log(\text{rate})$.
- The `market_index` feature drives the time pattern: daily mean rate per mile correlates 0.58 with `market_index`, and it shows a weekly sawtooth (rates rise through the week, dip on weekends).
- Equipment matters: Reefer (+13%) and Flatbed (+8%) price above Dry Van per mile. Heavier loads price slightly higher per mile.
- `quote_signal` carries only a weak, mostly incremental signal (row-level correlation with rate per mile ~ 0.06).
- The validation set introduces 8 cities never seen in training (e.g. Chicago, Charlotte, San Diego) and 1,461 loads on unseen routes - so the model leans on coordinates, not memorized lanes.



Data quality issues and how they were handled

- Outliers: 152 training rows (0.3%) have implausible rate-per-mile values ($< \$0.50/\text{mi}$ or $> \$10/\text{mi}$, e.g. a \$25,533 load). These were dropped from training; the L1 objective adds further robustness.
- Missing values: weight is missing for 300 rows and market_index for 374 (~0.7% each, spread evenly across months - consistent with missing at random). LightGBM handles NaNs natively, so no imputation was needed.
- Coordinates are constant per city, so they act as a geographic encoding and generalize to the 8 unseen validation cities.
- load_id values are unique and disjoint between train and validation; no leakage through identifiers.



Validation approach and train/test split

The development data (Jan-Oct 2025) was split by time, not randomly: training on January-September and validating on October - the month immediately before the November-December prediction window. This mirrors the real task, where the future must be predicted from the past. A random split would leak lane- and market-level patterns across the boundary and report over-optimistic metrics.

October holdout results (4,831 loads):

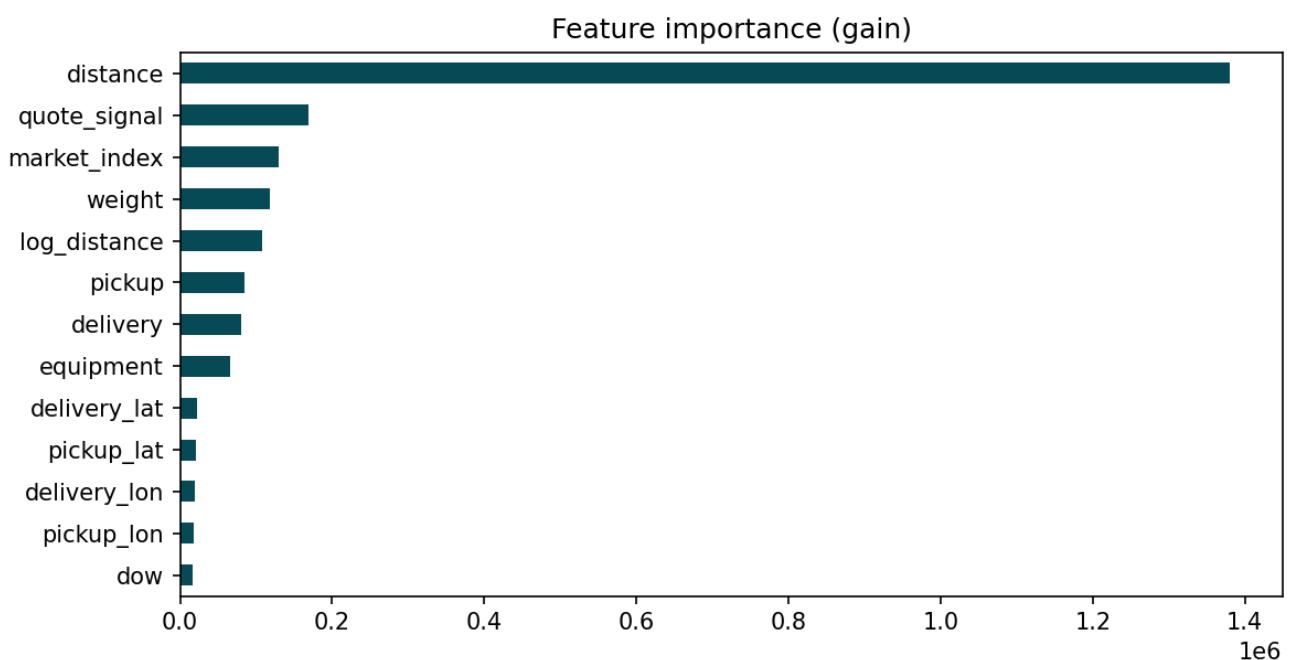
MAE	\$117.77
RMSE	\$612.74 (inflated by a handful of extreme loads)
MAPE	4.63%
Median APE	2.71%
R2	0.8371

Model selection (objective, target transform, feature set, tree size) was done exclusively on this holdout. The final model was then refit on all ten months with the early-stopped iteration count, using 5 seeds averaged at inference for stability.

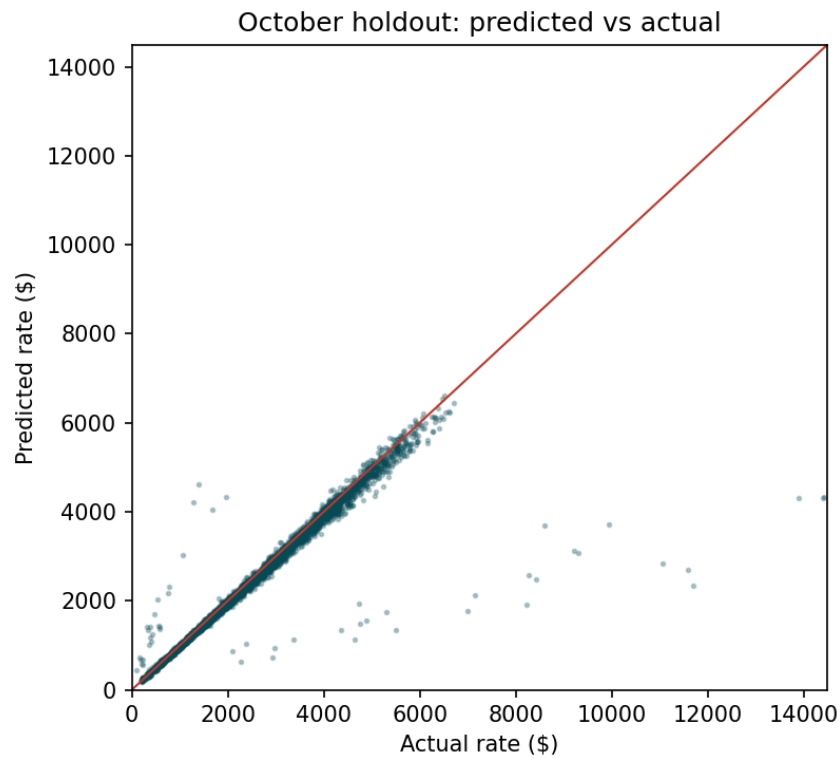
Model choice

Gradient-boosted trees (LightGBM) were chosen because the relationships are non-linear and full of interactions (lane x equipment x market), the data is tabular and mid-sized, native categorical support avoids fragile one-hot encoding of 64 cities, and NaNs are handled natively. Predicting $\log(\text{rate})$ with an L1 objective matched the multiplicative error structure and outperformed raw-target, L2, Huber and Tweedie variants on the holdout. The ~4,000-value route string was tested and dropped - it overfit relative to city + coordinate features. A month feature was also deliberately excluded: November/December never appear in training, and month was confirmed to hurt holdout performance; the model instead uses `market_index` plus `day-of-week` to capture time effects that generalize to unseen months.

Features: distance and $\log(\text{distance})$, weight, equipment, pickup/delivery city (categorical), pickup/delivery coordinates, `market_index`, `quote_signal`, and `day-of-week`.

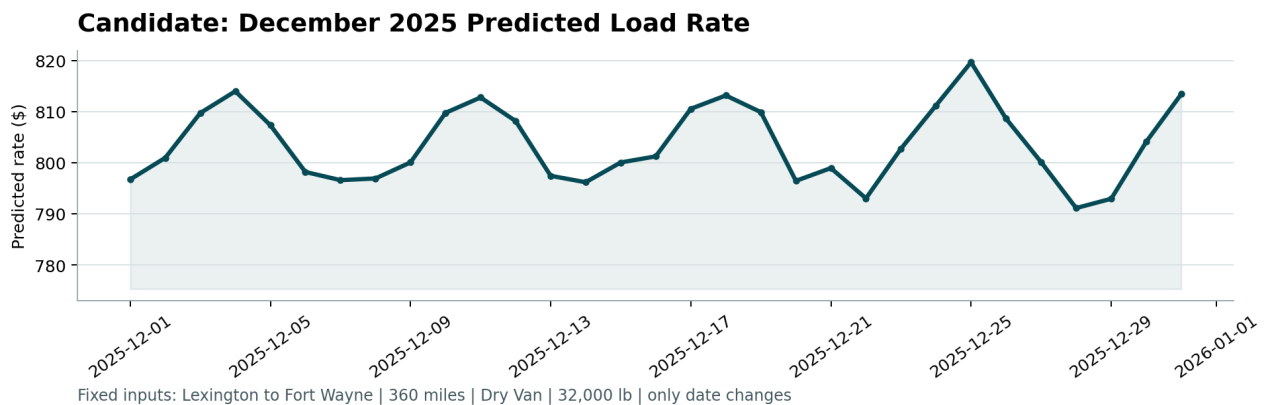


Holdout fit



December 2025 prediction chart (scorer output)

The December chart inputs contain no market_index, quote_signal or coordinates. Coordinates were attached from the city lookup; market_index and quote_signal were estimated per December date from the daily means observed in the validation set (which covers Dec 2025). The resulting curve reproduces the weekly sawtooth of market_index, with a Christmas-week peak - consistent with the route's history (32 training loads on this lane, mean rate \$857).



Reproduce

```
python -m pip install -r requirements.txt
python src/train.py      # holdout metrics + final models
python src/predict.py    # validation_predictions.csv + December inputs
python score.py --predictions validation_predictions.csv \
  --december-predictions data/december_chart_inputs.csv
```